

Singular value decomposition

- Review:
 - a) Definition: A quadratic form on $\mathbb{R}^n$ is a function Q whose value at $\mathbf{x}$ can be computed using the rule $Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \mathbf{x} \cdot \mathbf{A} \mathbf{x}$ where $\mathbf{A}$ is a symmetric matrix.
 - b) Minkowski 'inner product'. Write $\mathbb{R}^{n+1}$ as $(t, \mathbf{y})$ with $\mathbf{y} \in \mathbb{R}^n$.
 - 1) $Q((t, \mathbf{y})) = \|\mathbf{y}\|^2 - t^2$.
 - 2) The theory of special relativity is all about this Minkowski inner product.
 - c) Change of variable
 - 1) If $\mathbf{x} = \mathbf{P} \mathbf{y}$ where $\mathbf{P}$ is invertible, then $\mathbf{x}^T \mathbf{A} \mathbf{x} = \mathbf{y}^T \mathbf{P}^T \mathbf{A} \mathbf{P} \mathbf{y}$.
 - 2) Thus, in the new coordinates, Q is represented not by $\mathbf{A}$, but by $\mathbf{P}^T \mathbf{A} \mathbf{P}$.
 - d) Principle axis theorem: There is a change of variable matrix $\mathbf{P}$ that is orthonormal such that $Q(\mathbf{y}) = \mathbf{y}^T \mathbf{D} \mathbf{y}$ with $\mathbf{D}$ being diagonal (the eigenvalues of $\mathbf{A}$ are its diagonal entries).
 - e) Constrained optimization: Look at $Q(\mathbf{x})$ subject to the constraint that $\|\mathbf{x}\| = 1$.
 - 1) If $\mathbf{A}$ is the symmetric matrix such that $Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$, then the maximum of Q on the constraint surface is the largest eigenvalue of $\mathbf{A}$; and likewise the minimum is the smallest eigenvalue of $\mathbf{A}$.
 - 2) Minimize Q on the $\|\mathbf{x}\| = 1$ sphere subject to the constraint that $\mathbf{x}$ be orthogonal to all eigenvectors previously found.
- We looked at symmetric matrices: Let $\mathbf{A}$ be a symmetric $n \times n$ matrix.
 - a) The matrix $\mathbf{A}$ is orthogonally diagonalizable: $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$ with $\mathbf{P}^{-1} = \mathbf{P}^T$.
 - b) The columns of $\mathbf{P}$ are an orthonormal basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of eigenvectors for $\mathbf{A}$.
 - c) The eigenvalues of $\mathbf{A}$ are real and are the diagonal entries of $\mathbf{D}$.
- What can be said about an $m \times n$ matrix $\mathbf{A}$ when $m \neq n$?
 - a) $\mathbf{A}^T \mathbf{A}$ is a symmetric, $n \times n$ matrix. (We used this matrix $\mathbf{A}^T \mathbf{A}$ when doing least squares and orthogonal projection to a subspace in $\mathbb{R}^n$.)
 - b) Thus, it is orthogonally diagonalizable: $\mathbf{A}^T \mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$ with $\mathbf{P}^{-1} = \mathbf{P}^T$.
 - c) The eigenvalues are non-negative because $\mathbf{v} \cdot \mathbf{A}^T \mathbf{A} \mathbf{v} = (\mathbf{A} \mathbf{v}) \cdot (\mathbf{A} \mathbf{v}) = \|\mathbf{A} \mathbf{v}\|^2$.
 - d) The square roots of the eigenvalues $\sigma_1 = \sqrt{\lambda_1}, \dots, \sigma_n = \sqrt{\lambda_n}$ are called the singular values.
 - e) $\text{Null}(\mathbf{A})$ has dimension at least $n - m$, so at most m of the λ_k 's are not zero.

- We can likewise look at AA^T which is an $m \times m$, symmetric matrix.
 - a) It can be written as QBQ^{-1} where Q is an $m \times m$ orthogonal matrix and B is diagonal with entries being the eigenvalues of AA^T .
- Theorem: The non-zero eigenvalues of AA^T are the same as those of $A^T A$.
 - a) Indeed, if $A^T A \mathbf{v} = \lambda \mathbf{v}$, then $A(A^T A \mathbf{v}) = \lambda A \mathbf{v}$ and $A(A^T A \mathbf{v})$ is $(AA^T)A \mathbf{v}$.
 - b) Also, if $AA^T \mathbf{u} = \lambda' \mathbf{u}$, then $A^T(AA^T \mathbf{u}) = \lambda' A^T \mathbf{u}$.
 - c) So if $\mathbf{v}_k$ is an eigenvector of $A^T A$ with non-zero eigenvalue λ_k and length 1, then $\mathbf{u}_k = \lambda_k^{-1/2} A \mathbf{v}_k$ is an eigenvector of AA^T with eigenvalue λ_k and length 1.
 - d) Moreover $\mathbf{u}_k \cdot \mathbf{u}_j = \lambda_k^{-1/2} \lambda_j^{-1/2} (A \mathbf{v}_k) \cdot (A \mathbf{v}_j) = \lambda_k^{-1/2} \lambda_j^{-1/2} \mathbf{v}_k \cdot A^T A \mathbf{v}_j = \lambda_k^{-1/2} \lambda_j^{1/2} \mathbf{v}_k \cdot \mathbf{v}_j$.
 - e) Thus, the $\{\mathbf{u}_k = \sigma_k^{-1} A \mathbf{v}_k : \lambda_k \neq 0\}$ are an orthonormal basis for the eigenspaces of AA^T with non-zero eigenvalue.
- Thus, AA^T and $A^T A$ have the same set of non-zero eigenvalues (which are non-negative). List these as $\lambda_1, \dots, \lambda_r$ for $r \leq \min(m, n)$.
- I can write $A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T$ where $\{\mathbf{v}_i\}_{i=1, \dots, r}$ are an orthonormal basis for the eigenspaces of A with non-zero eigenvalue, and $\mathbf{u}_k = \sigma_k^{-1} A \mathbf{v}_k$.
 - a) Indeed, if $\mathbf{x}$ is any vector in $\mathbb{R}^n$, I can use the basis $\{\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}$ to write $\mathbf{x}$ as $\mathbf{x} = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n$.
 - b) Then $A \mathbf{x} = \sigma_1 c_1 \mathbf{u}_1 + \dots + \sigma_r c_r \mathbf{u}_r$ which gives the expansion of $A \mathbf{x}$ in terms of the $\mathbf{u}$ -basis. (Note that if $AA^T \mathbf{u} = 0$, then $A^T \mathbf{u} = 0$ so $A \mathbf{x}$ has no $\mathbf{u}_k$'s in its expansion from the 0 eigenvalues of AA^T .) Likewise, there are no c_k for those $\mathbf{v}_k$ with $A \mathbf{v}_k$ being 0.
 - c) In this regards, $\mathbf{u}_k \cdot A \mathbf{x} = (A^T \mathbf{u}_k) \cdot \mathbf{x} = \sigma_k \mathbf{v}_k \cdot \mathbf{x} = \sigma_k c_k$ as it should be.
- The depiction of A as $\sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T$ is called the *singular value decomposition* of A .
- Arc of Math 22a:
 - a) We started with linear equations, wrote them in terms of matrices ($A \mathbf{x} = \mathbf{b}$) and then introduced the row-reduction algorithm to solve them or find them inconsistent by obtaining $\text{rref}(A)$.
 - b) We then started playing with the notion of a matrix:
 - 1) Multiplying them against vectors, multiplying them against other matrices.
 - 2) The notion of an invertible matrix and its inverse.
 - 3) The set of invertible matrices form a group denoted by $\text{Gl}(n; \mathbb{R})$.
 - 4) Introduced elementary matrices to formalize the row reduction operation.
 - c) Then, we discovered the determinant of a matrix which gives as a way to determine when a matrix is invertible.

- 1) Lots of neat properties of determinants ($\det(AB) = \det(A)\det(B)$).
 - 2) An inductive formula for the determinant using expansion across any row or column.
 - d) Introduced the notion of a vector space
 - 1) Subspaces in $\mathbb{R}^n$.
 - 2) Null space and column space of a matrix.
 - 3) The notion of a basis for a vector space giving coordinates for the vector space. which allows us to write linear transformations as matrix multiplication.
 - 4) Change of basis formulas.
 - 5) Notion of dimension of a subspace and rank of a matrix (dimension of its column space).
 - e) The notion of an eigenvector and eigenvalue for a square matrix.
 - 1) Characteristic equation for potential eigenvalues.
 - 2) Diagonalizable matrices (not all are diagonalizable over $\mathbb{R}$).
 - 3) Complex matrices and complex eigenvalues (most matrices are diagonalizable if complex eigenvalues are allowed).
 - 4) Using diagonalization to solve discrete dynamical systems and linear differential equations.
 - f) Introduced dot products (inner products) and studied the many things it brings with it.
 - 1) Notion of length, distance, orthogonality.
 - 2) Orthogonal basis, orthonormal basis, orthogonal projections
 - 3) Gram-Schmidt algorithm
 - 4) Least squares and finding closest vector in a subspace.
 - g) Continuing with dot product, we studied symmetric matrices:
 - 1) These are orthogonally diagonalizable (and vice versa).
 - 2) There is an orthonormal basis consisting of eigenvectors. Eigenvalues are real.
 - 3) Spectral theorem
 - 4) Singular value decomposition for $m \times n$ matrices even when $m \neq n$.
- And we learned how to write proofs and interpret rigorous mathematics.
 - And we learned something about science and data.